## APPENDIX A
## DETAILED NUMERICAL RESULTS OF NOWCASTING EFFECTIVENESS

Tables IV and V report the detailed numerical values used for Figures 2 and 3 in the main paper, respectively. We report the mean and standard deviation over three runs with different random seeds. As mentioned in Section V-B, MULTIAIR outperforms all baseline methods across all lead times, in terms of both classification and regression metrics.

## APPENDIX B
## DETAILED NUMERICAL RESULTS OF ABLATION STUDY

Tables VII and VIII report the detailed numerical results of the ablation study in terms of classification and regression metrics, respectively. As discussed in Section V-C, MULTIAIR achieves the best overall performance, showing the effectiveness of its key components.

## APPENDIX C
## HANDLING OF MISSING VALUES

In the case of time series data, missing values at each station $s \in S_T \cup S_C$ at $t' \in \{t - T_t, ..., t\}$ are filled with valid observations from the nearest station at $t'$. In this case, to distinguish between actual observed values and imputed values, a missing value indicator $\boldsymbol{m}_s^{(t')} \in \{0, 1\}^{d_t}$ is introduced, where $d_t$ denotes the number of observed values in a ground observation sensor. This multi-hot vector indicates which values among $PM_{2.5}$ value and other observation values are originally missing for each station $s$ at time $t'$ and is used in constructing the input time series data. For AOD images, missing values do not occur independently like ground observations but can appear simultaneously over a wide area, making the proposed imputation method inaccurate. Thus, we fill the missing values with -1, as it explicitly falls outside the typical AOD range (about 0 to 4).

## APPENDIX D
## ADDITIONAL EXPERIMENTAL SETTINGS

In this section, we introduce detailed settings for training MULTIAIR and baseline methods in Section V-A, including hyperparameters.

For pre-training image models DeepRaNE and SimVP, we followed the training scheme of ASOC+ [16]. We used Mean Squared Error (MSE) loss and the Adam [31] optimizer with a learning rate $2 \times 10^{-5}$. We set the batch size to 20 and the number of training steps to 50,000. For every 1,000 steps, we evaluated the model by calculating loss on the validation set and selected the model with the best performance. The number of hidden channels of DeepRaNE was set to 32.

We reproduced the architectures of CrossViViT and GNN-LSTM from [15] and [14], respectively, and adapted their implementations to match our experimental settings. Our task requires station-wise multivariate forecasting over multiple monitoring stations; therefore, we adopt CrossViViT-MS, a station-wise multi-station adaptation of CrossViViT, as a multimodal baseline. For GNN-LSTM, we constructed graphs with stations as nodes, edges between stations, and the speed and direction of wind as the edge weights. To ensure a fair comparison with our method, we used only the ground-based observations on air quality and AOD image. We used Mean Squared Error (MSE) loss and the Adam [31] optimizer with a learning rate $10^{-3}$ for CrossViViT, AirFormer, and GNN-LSTM.

We used the modified focal-R L1 loss in Section IV-D with hyperparameters $\beta = 0.25$ and $\gamma = 2$ and used the Adam optimizer with a learning rate $10^{-3}$ for all other training processes. We set the batch size to 20 for ASOC+ and 120 for other models. We set the maximum number of epochs to 200 with early stopping if there was no performance gain for 5 epochs. For ASOC+, we set the hidden dimension for LSTM cells to 64 and the number of heads for the attention blocks to 4. For MULTIAIR, we set the hidden dimension for LSTM cells to 128.

For all time-series forecasting baseline methods, we follow their original hyperparameter settings without additional tuning. Specifically, the Transformer uses a model dimension of 64 with 8 attention heads and 2 layers. TimesNet uses 2 TimesBlocks with a feed-forward dimension of 32, while TimeMixer uses 2 PDM blocks with a feed-forward dimension of 32.

## APPENDIX E
## EVALUATION USING EXTRA METRICS

The classification metrics used in the main paper, the critical success index (CSI) and F1 score, are defined for each cumulative class $c$ as:

$$\text{CSI}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad \text{F1}_c = \frac{2 \times \text{TP}_c}{2 \times \text{TP}_c + \text{FP}_c + \text{FN}_c},$$

where $\text{TP}_c, \text{FP}_c, \text{FN}_c$ stand for the counts of true positives, false positives, and false negatives, respectively, for each cumulative class $c$.

In this section, we also introduce evaluation results using extra evaluation metrics. For extra evaluation metrics, we use four classification metrics: accuracy (ACC), probability of detection (POD), false alarm rate (FAR), and F1 score, and three regression metrics: RMSE, normalized mean error (NME), and Pearson correlation coefficient ($\rho$). For extra evaluation metrics, we simply evaluate once for all prediction lead times. ACC is defined as the ratio of correct class prediction based on 4 classes in Section V-A: *Severity 1, 2, 3* and *4*. Other classification metrics, POD, FAR, and F1 score, focus on predicting extreme air pollution of *Severity 3 or worse*, which are defined as follows:

$$\text{POD} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FAR} = \frac{\text{FP}}{\text{TP} + \text{FP}},$$
$$\text{F1} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}},$$

TABLE IV

PERFORMANCE ON THE TEST SET EVALUATED USING CLASSIFICATION METRICS (CSI AND F1 SCORES, WHERE HIGHER VALUES INDICATE BETTER PERFORMANCE). RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE RANDOM SEEDS. MULTIAIR PERFORMS BEST ACROSS ALL POLLUTION LEVELS AND LEAD TIMES. IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

| Metric | Lead time | ASOC+ | CrossViViT-MS | LSTM | Transformer | TimesNet | TimeMixer | AirFormer | GNN-LSTM | **MULTIAIR** |
|---|---|---|---|---|---|---|---|---|---|---|
| CSI score Sev. 2 or worse ($> 15\mu g/m^3$) | 1 | 0.721 ± 0.003 | 0.795 ± 0.014 | 0.797 ± 0.006 | 0.743 ± 0.043 | 0.685 ± 0.011 | 0.787 ± 0.003 | 0.755 ± 0.011 | 0.805 ± 0.003 | **0.822** ± 0.000 |
| | 2 | 0.705 ± 0.004 | 0.752 ± 0.008 | 0.711 ± 0.035 | 0.675 ± 0.011 | 0.742 ± 0.005 | 0.749 ± 0.005 | 0.704 ± 0.015 | 0.758 ± 0.005 | **0.786** ± 0.000 |
| | 3 | 0.687 ± 0.004 | 0.719 ± 0.005 | 0.711 ± 0.003 | 0.683 ± 0.028 | 0.665 ± 0.012 | 0.697 ± 0.012 | 0.666 ± 0.024 | 0.720 ± 0.006 | **0.757** ± 0.000 |
| | 4 | 0.667 ± 0.004 | 0.693 ± 0.002 | 0.681 ± 0.003 | 0.660 ± 0.023 | 0.652 ± 0.012 | 0.678 ± 0.006 | 0.631 ± 0.032 | 0.688 ± 0.007 | **0.732** ± 0.001 |
| | 5 | 0.647 ± 0.004 | 0.671 ± 0.001 | 0.655 ± 0.002 | 0.641 ± 0.017 | 0.641 ± 0.012 | 0.645 ± 0.014 | 0.601 ± 0.038 | 0.660 ± 0.008 | **0.709** ± 0.001 |
| | 6 | 0.628 ± 0.004 | 0.653 ± 0.001 | 0.632 ± 0.002 | 0.623 ± 0.013 | 0.630 ± 0.011 | 0.637 ± 0.008 | 0.576 ± 0.047 | 0.635 ± 0.010 | **0.689** ± 0.001 |
| | Avg. | 0.676 ± 0.004 | 0.714 ± 0.005 | 0.704 ± 0.004 | 0.677 ± 0.027 | 0.658 ± 0.011 | 0.698 ± 0.008 | 0.656 ± 0.028 | 0.711 ± 0.006 | **0.749** ± 0.000 |
| F1 score Sev. 2 or worse ($> 15\mu g/m^3$) | 1 | 0.838 ± 0.002 | 0.886 ± 0.009 | 0.887 ± 0.004 | 0.852 ± 0.028 | 0.813 ± 0.008 | 0.881 ± 0.002 | 0.861 ± 0.007 | 0.892 ± 0.002 | **0.902** ± 0.000 |
| | 2 | 0.827 ± 0.003 | 0.859 ± 0.005 | 0.857 ± 0.003 | 0.831 ± 0.024 | 0.806 ± 0.008 | 0.852 ± 0.003 | 0.826 ± 0.010 | 0.862 ± 0.003 | **0.880** ± 0.000 |
| | 3 | 0.814 ± 0.003 | 0.837 ± 0.003 | 0.831 ± 0.002 | 0.812 ± 0.020 | 0.799 ± 0.008 | 0.821 ± 0.008 | 0.799 ± 0.017 | 0.837 ± 0.004 | **0.862** ± 0.000 |
| | 4 | 0.800 ± 0.003 | 0.819 ± 0.001 | 0.810 ± 0.002 | 0.795 ± 0.016 | 0.789 ± 0.009 | 0.808 ± 0.004 | 0.774 ± 0.025 | 0.815 ± 0.005 | **0.845** ± 0.000 |
| | 5 | 0.786 ± 0.003 | 0.803 ± 0.000 | 0.791 ± 0.001 | 0.781 ± 0.013 | 0.781 ± 0.009 | 0.784 ± 0.010 | 0.751 ± 0.030 | 0.795 ± 0.006 | **0.830** ± 0.001 |
| | 6 | 0.771 ± 0.003 | 0.790 ± 0.001 | 0.775 ± 0.002 | 0.768 ± 0.010 | 0.773 ± 0.008 | 0.778 ± 0.006 | 0.730 ± 0.038 | 0.777 ± 0.007 | **0.816** ± 0.001 |
| | Avg. | 0.806 ± 0.003 | 0.832 ± 0.003 | 0.825 ± 0.002 | 0.806 ± 0.018 | 0.793 ± 0.008 | 0.821 ± 0.006 | 0.790 ± 0.021 | 0.830 ± 0.004 | **0.856** ± 0.000 |
| CSI score Sev. 3 or worse ($> 35\mu g/m^3$) | 1 | 0.568 ± 0.011 | 0.701 ± 0.014 | 0.710 ± 0.013 | 0.562 ± 0.093 | 0.550 ± 0.004 | 0.695 ± 0.005 | 0.694 ± 0.031 | 0.709 ± 0.004 | **0.755** ± 0.001 |
| | 2 | 0.542 ± 0.012 | 0.603 ± 0.010 | 0.601 ± 0.020 | 0.514 ± 0.080 | 0.530 ± 0.003 | 0.599 ± 0.013 | 0.606 ± 0.028 | 0.609 ± 0.001 | **0.684** ± 0.002 |
| | 3 | 0.514 ± 0.011 | 0.523 ± 0.011 | 0.512 ± 0.032 | 0.474 ± 0.068 | 0.505 ± 0.009 | 0.511 ± 0.021 | 0.542 ± 0.028 | 0.541 ± 0.003 | **0.631** ± 0.002 |
| | 4 | 0.486 ± 0.011 | 0.463 ± 0.009 | 0.441 ± 0.044 | 0.440 ± 0.057 | 0.475 ± 0.014 | 0.470 ± 0.015 | 0.484 ± 0.028 | 0.492 ± 0.004 | **0.588** ± 0.002 |
| | 5 | 0.458 ± 0.011 | 0.414 ± 0.010 | 0.375 ± 0.062 | 0.411 ± 0.048 | 0.450 ± 0.016 | 0.414 ± 0.025 | 0.432 ± 0.031 | 0.451 ± 0.006 | **0.551** ± 0.002 |
| | 6 | 0.433 ± 0.011 | 0.377 ± 0.009 | 0.324 ± 0.078 | 0.387 ± 0.041 | 0.429 ± 0.014 | 0.381 ± 0.017 | 0.386 ± 0.038 | 0.416 ± 0.007 | **0.519** ± 0.001 |
| | Avg. | 0.500 ± 0.011 | 0.513 ± 0.010 | 0.494 ± 0.0415 | 0.465 ± 0.065 | 0.490 ± 0.010 | 0.512 ± 0.016 | 0.524 ± 0.031 | 0.536 ± 0.004 | **0.621** ± 0.001 |
| F1 score Sev. 3 or worse ($> 35\mu g/m^3$) | 1 | 0.725 ± 0.009 | 0.824 ± 0.009 | 0.831 ± 0.009 | 0.717 ± 0.074 | 0.710 ± 0.003 | 0.820 ± 0.003 | 0.819 ± 0.022 | 0.830 ± 0.003 | **0.861** ± 0.001 |
| | 2 | 0.703 ± 0.010 | 0.752 ± 0.008 | 0.751 ± 0.016 | 0.677 ± 0.068 | 0.693 ± 0.003 | 0.749 ± 0.010 | 0.755 ± 0.021 | 0.757 ± 0.001 | **0.812** ± 0.001 |
| | 3 | 0.679 ± 0.010 | 0.687 ± 0.009 | 0.677 ± 0.027 | 0.641 ± 0.061 | 0.671 ± 0.008 | 0.676 ± 0.018 | 0.703 ± 0.024 | 0.702 ± 0.002 | **0.774** ± 0.001 |
| | 4 | 0.654 ± 0.010 | 0.632 ± 0.009 | 0.611 ± 0.042 | 0.610 ± 0.054 | 0.644 ± 0.013 | 0.639 ± 0.014 | 0.652 ± 0.026 | 0.659 ± 0.004 | **0.741** ± 0.001 |
| | 5 | 0.628 ± 0.010 | 0.585 ± 0.010 | 0.544 ± 0.065 | 0.582 ± 0.048 | 0.620 ± 0.015 | 0.585 ± 0.025 | 0.603 ± 0.030 | 0.622 ± 0.006 | **0.711** ± 0.001 |
| | 6 | 0.604 ± 0.011 | 0.548 ± 0.010 | 0.486 ± 0.087 | 0.527 ± 0.042 | 0.601 ± 0.013 | 0.551 ± 0.017 | 0.556 ± 0.040 | 0.587 ± 0.007 | **0.683** ± 0.001 |
| | Avg. | 0.666 ± 0.010 | 0.671 ± 0.009 | 0.650 ± 0.041 | 0.631 ± 0.058 | 0.656 ± 0.009 | 0.670 ± 0.015 | 0.681 ± 0.027 | 0.693 ± 0.004 | **0.764** ± 0.001 |
| CSI score Sev. 4 ($> 75\mu g/m^3$) | 1 | 0.361 ± 0.036 | 0.459 ± 0.205 | 0.561 ± 0.071 | 0.324 ± 0.190 | 0.308 ± 0.106 | 0.590 ± 0.009 | 0.489 ± 0.159 | 0.558 ± 0.011 | **0.696** ± 0.005 |
| | 2 | 0.322 ± 0.031 | 0.337 ± 0.167 | 0.382 ± 0.094 | 0.279 ± 0.166 | 0.273 ± 0.105 | 0.429 ± 0.012 | 0.389 ± 0.125 | 0.426 ± 0.014 | **0.593** ± 0.008 |
| | 3 | 0.279 ± 0.025 | 0.246 ± 0.136 | 0.257 ± 0.112 | 0.242 ± 0.146 | 0.238 ± 0.108 | 0.316 ± 0.026 | 0.314 ± 0.114 | 0.328 ± 0.024 | **0.518** ± 0.009 |
| | 4 | 0.242 ± 0.021 | 0.188 ± 0.114 | 0.182 ± 0.118 | 0.212 ± 0.130 | 0.202 ± 0.101 | 0.243 ± 0.009 | 0.248 ± 0.109 | 0.263 ± 0.026 | **0.460** ± 0.011 |
| | 5 | 0.208 ± 0.019 | 0.148 ± 0.096 | 0.133 ± 0.113 | 0.187 ± 0.114 | 0.172 ± 0.095 | 0.188 ± 0.032 | 0.189 ± 0.096 | 0.211 ± 0.029 | **0.407** ± 0.012 |
| | 6 | 0.178 ± 0.019 | 0.118 ± 0.079 | 0.097 ± 0.094 | 0.165 ± 0.101 | 0.155 ± 0.846 | 0.141 ± 0.022 | 0.144 ± 0.075 | 0.171 ± 0.030 | **0.358** ± 0.011 |
| | Avg. | 0.265 ± 0.025 | 0.249 ± 0.133 | 0.269 ± 0.100 | 0.235 ± 0.141 | 0.225 ± 0.100 | 0.318 ± 0.018 | 0.295 ± 0.113 | 0.326 ± 0.022 | **0.505** ± 0.009 |
| F1 score Sev. 4 ($> 75\mu g/m^3$) | 1 | 0.530 ± 0.039 | 0.610 ± 0.212 | 0.717 ± 0.057 | 0.470 ± 0.203 | 0.464 ± 0.130 | 0.742 ± 0.007 | 0.646 ± 0.153 | 0.716 ± 0.009 | **0.821** ± 0.004 |
| | 2 | 0.487 ± 0.036 | 0.488 ± 0.201 | 0.548 ± 0.095 | 0.420 ± 0.191 | 0.420 ± 0.136 | 0.601 ± 0.011 | 0.552 ± 0.137 | 0.597 ± 0.014 | **0.745** ± 0.006 |
| | 3 | 0.436 ± 0.031 | 0.381 ± 0.185 | 0.401 ± 0.136 | 0.376 ± 0.179 | 0.376 ± 0.149 | 0.480 ± 0.030 | 0.470 ± 0.139 | 0.494 ± 0.027 | **0.683** ± 0.008 |
| | 4 | 0.389 ± 0.028 | 0.306 ± 0.170 | 0.297 ± 0.162 | 0.338 ± 0.167 | 0.327 ± 0.148 | 0.391 ± 0.011 | 0.388 ± 0.146 | 0.416 ± 0.033 | **0.630** ± 0.011 |
| | 5 | 0.344 ± 0.026 | 0.249 ± 0.152 | 0.224 ± 0.168 | 0.305 ± 0.154 | 0.286 ± 0.146 | 0.316 ± 0.045 | 0.310 ± 0.140 | 0.347 ± 0.038 | **0.579** ± 0.012 |
| | 6 | 0.302 ± 0.027 | 0.205 ± 0.132 | 0.169 ± 0.150 | 0.276 ± 0.142 | 0.262 ± 0.132 | 0.247 ± 0.033 | 0.247 ± 0.117 | 0.291 ± 0.043 | **0.528** ± 0.012 |
| | Avg. | 0.415 ± 0.031 | 0.373 ± 0.175 | 0.393 ± 0.128 | 0.364 ± 0.173 | 0.356 ± 0.140 | 0.463 ± 0.023 | 0.435 ± 0.139 | 0.477 ± 0.027 | **0.664** ± 0.009 |

where TP, FP, FN stand for the counts of true positives, false positives, and false negatives, respectively, for air pollution of *Severity 3 or worse*. NME, $\rho$ is defined as:

$$\text{NME} = \frac{\sum |\hat{\mathbf{P}}_T - \mathbf{P}_T|}{\sum \mathbf{P}_T} \times 100,$$

$$\rho = \frac{\sum((\hat{\mathbf{P}}_T - \mu_{\hat{\mathbf{P}}_T}) \times (\mathbf{P}_T - \mu_{\mathbf{P}_T}))}{\sqrt{\sum(\hat{\mathbf{P}}_T - \mu_{\hat{\mathbf{P}}_T})^2 \times \sum(\mathbf{P}_T - \mu_{\mathbf{P}_T})^2}},$$

where $\mathbf{P}_T, \hat{\mathbf{P}}_T$ each denotes all ground truth $PM_{2.5}$ values and predicted $PM_{2.5}$ values respectively, and $\mu_{\mathbf{P}_T}$, $\mu_{\hat{\mathbf{P}}_T}$ each denotes the mean value of ground truth $PM_{2.5}$ values and predicted $PM_{2.5}$ values respectively.

Table IV, Table V and Table VI present the evaluation results in comparison with the baseline models. MULTIAIR performs the best for every metric except for FAR. However, AirFormer, which achieves the best FAR, records the lowest POD and F1 score among the models, suggesting that it simply underestimates severe air pollution.

Table VII, Table VIII and Table IX present the evaluation results in comparison with the ablation models, which are described in Section V-C. MULTIAIR achieves the best or second-best performance for all metrics. Overall, MULTIAIR provides strong and balanced performance for extreme air pollution nowcasting.

## APPENDIX F
## HYPERPARAMETER SENSITIVITY

In this section, we evaluate the sensitivity of several hyperparameters in MULTIAIR, including the input time-series length, the number of hidden dimensions, and the focal loss parameter $\gamma$. As shown in Table X, increasing the input time-series length generally improves performance, although the gains are marginal. Similarly, the results indicate that MULTIAIR is relatively insensitive to the choice of hidden dimension

TABLE V

PERFORMANCE ON THE TEST SET EVALUATED USING REGRESSION METRICS (RMSE AND MAE, WHERE LOWER VALUES INDICATE BETTER PERFORMANCE). RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE RANDOM SEEDS. MULTIAIR PERFORMS BEST ACROSS ALL LEAD TIMES. IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

| Metric | Lead time | ASOC+ | CrossViViT-MS | LSTM | Transformer | TimesNet | TimeMixer | AirFormer | GNN-LSTM | **MULTIAIR** |
|---|---|---|---|---|---|---|---|---|---|---|
| RMSE | 1 | 4.944 ± 0.012 | 5.121 ± 0.204 | 4.826 ± 0.210 | 7.055 ± 1.479 | 7.713 ± 0.061 | 5.081 ± 0.103 | 5.470 ± 0.166 | 5.126 ± 0.055 | **4.008** ± 0.013 |
| | 2 | 6.795 ± 0.015 | 6.615 ± 0.071 | 6.480 ± 0.251 | 7.921 ± 1.236 | 8.097 ± 0.034 | 6.601 ± 0.167 | 6.735 ± 0.147 | 6.581 ± 0.018 | **5.201** ± 0.027 |
| | 3 | 8.132 ± 0.016 | 7.790 ± 0.071 | 7.761 ± 0.337 | 8.669 ± 1.020 | 8.545 ± 0.005 | 7.910 ± 0.242 | 7.693 ± 0.269 | 7.695 ± 0.071 | **6.134** ± 0.033 |
| | 4 | 9.176 ± 0.016 | 8.688 ± 0.164 | 8.763 ± 0.428 | 9.319 ± 0.825 | 9.105 ± 0.044 | 8.600 ± 0.190 | 8.588 ± 0.352 | 8.593 ± 0.112 | **6.920** ± 0.041 |
| | 5 | 10.033 ± 0.015 | 9.404 ± 0.248 | 9.606 ± 0.527 | 9.893 ± 0.647 | 9.615 ± 0.101 | 9.444 ± 0.199 | 9.301 ± 0.391 | 9.378 ± 0.137 | **7.613** ± 0.050 |
| | 6 | 10.734 ± 0.008 | 9.949 ± 0.295 | 10.288 ± 0.613 | 10.394 ± 0.498 | 10.071 ± 0.229 | 9.850 ± 0.191 | 9.939 ± 0.499 | 10.077 ± 0.149 | **8.231** ± 0.067 |
| | Avg. | 8.303 ± 0.014 | 7.928 ± 0.175 | 7.954 ± 0.394 | 8.875 ± 0.951 | 8.858 ± 0.079 | 7.914 ± 0.182 | 7.949 ± 0.304 | 7.908 ± 0.090 | **6.351** ± 0.037 |
| MAE | 1 | 3.253 ± 0.001 | 3.332 ± 0.109 | 3.221 ± 0.149 | 4.498 ± 0.903 | 5.381 ± 0.145 | 3.402 ± 0.062 | 3.905 ± 0.153 | 3.409 ± 0.008 | **2.728** ± 0.007 |
| | 2 | 4.394 ± 0.002 | 4.249 ± 0.071 | 4.214 ± 0.163 | 5.085 ± 0.792 | 5.618 ± 0.132 | 4.352 ± 0.103 | 4.672 ± 0.176 | 4.308 ± 0.045 | **3.445** ± 0.011 |
| | 3 | 5.228 ± 0.003 | 4.964 ± 0.044 | 5.010 ± 0.217 | 5.591 ± 0.688 | 5.893 ± 0.154 | 5.198 ± 0.186 | 5.233 ± 0.252 | 5.010 ± 0.086 | **4.000** ± 0.014 |
| | 4 | 5.896 ± 0.004 | 5.524 ± 0.022 | 5.647 ± 0.278 | 6.036 ± 0.586 | 6.245 ± 0.211 | 5.691 ± 0.115 | 5.773 ± 0.320 | 5.586 ± 0.124 | **4.474** ± 0.017 |
| | 5 | 6.454 ± 0.004 | 5.983 ± 0.027 | 6.178 ± 0.331 | 6.434 ± 0.490 | 6.568 ± 0.253 | 6.264 ± 0.151 | 6.232 ± 0.339 | 6.099 ± 0.160 | **4.898** ± 0.018 |
| | 6 | 6.917 ± 0.002 | 6.354 ± 0.042 | 6.607 ± 0.369 | 6.785 ± 0.409 | 6.884 ± 0.333 | 6.506 ± 0.137 | 6.631 ± 0.427 | 6.569 ± 0.196 | **5.277** ± 0.024 |
| | Avg. | 5.357 ± 0.003 | 5.068 ± 0.053 | 5.146 ± 0.251 | 5.738 ± 0.645 | 6.098 ± 0.205 | 5.236 ± 0.126 | 5.408 ± 0.278 | 5.163 ± 0.103 | **4.137** ± 0.015 |

TABLE VI

PERFORMANCE ON THE TEST SET IN TERMS OF EXTRA METRICS. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE RANDOM SEEDS. FOR METRICS WITH AN UPWARD ARROW (↑), HIGHER VALUES INDICATE BETTER PERFORMANCE, WHEREAS FOR THOSE WITH A DOWNWARD ARROW (↓), LOWER VALUES ARE PREFERABLE. IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

| Metric | | ASOC+ | CrossViViT-MS | LSTM | Transformer | TimesNet | TimeMixer | AirFormer | GNN-LSTM | **MULTIAIR** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Class.** | ACC (↑) | 0.747 ± 0.003 | 0.766 ± 0.010 | 0.766 ± 0.010 | 0.740 ± 0.028 | 0.713 ± 0.011 | 0.759 ± 0.005 | 0.765 ± 0.011 | 0.774 ± 0.010 | **0.810** ± 0.001 |
| | POD (↑) | 0.600 ± 0.021 | 0.601 ± 0.003 | 0.568 ± 0.051 | 0.564 ± 0.081 | 0.643 ± 0.049 | 0.604 ± 0.019 | 0.573 ± 0.045 | 0.653 ± 0.009 | **0.733** ± 0.006 |
| | FAR (↓) | 0.250 ± 0.009 | 0.224 ± 0.028 | 0.212 ± 0.007 | 0.280 ± 0.046 | 0.327 ± 0.039 | 0.234 ± 0.002 | **0.139** ± 0.030 | 0.258 ± 0.009 | 0.200 ± 0.008 |
| | F1 (↑) | 0.666 ± 0.010 | 0.677 ± 0.009 | 0.659 ± 0.036 | 0.631 ± 0.058 | 0.656 ± 0.008 | 0.676 ± 0.013 | 0.687 ± 0.024 | 0.695 ± 0.003 | **0.765** ± 0.001 |
| **Reg.** | RMSE (↓) | 8.392 ± 0.016 | 8.099 ± 0.117 | 8.168 ± 0.410 | 8.956 ± 0.890 | 8.896 ± 0.072 | 8.083 ± 0.171 | 8.093 ± 0.313 | 8.040 ± 0.083 | **6.510** ± 0.040 |
| | NME (↓) | 29.439 ± 0.269 | 27.237 ± 0.176 | 27.665 ± 1.333 | 30.850 ± 3.456 | 32.786 ± 1.088 | 28.146 ± 0.618 | 29.072 ± 1.480 | 27.625 ± 0.519 | **22.240** ± 0.080 |
| | $\rho$ (↑) | 0.826 ± 0.001 | 0.842 ± 0.008 | 0.838 ± 0.016 | 0.796 ± 0.043 | 0.804 ± 0.002 | 0.838 ± 0.007 | 0.875 ± 0.001 | 0.840 ± 0.004 | **0.900** ± 0.002 |

size and the value of $\gamma$, exhibiting stable performance across a diverse range of hyperparameter settings.

## APPENDIX G
## ANALYSIS OF THE ERROR-ADAPTIVE LOSS

To examine the motivation for the error-adaptive loss in Eq. (11), we evaluate MULTIAIR at random initialization and after training, and compare the mean focal-R loss across observed $PM_{2.5}$ concentrations. As shown in Fig. 5, the mean loss generally increases with concentration on both the training and test sets, with larger losses in the Severity 3 and 4 ranges. The trained model yields lower losses across the concentration range, with particularly large absolute reductions at high concentrations. These observations indicate larger per-sample loss contributions from high-pollution cases and support the rationale for using error-adaptive weighting to emphasize difficult observations.

TABLE VII

ABLATION STUDY RESULTS BASED ON CLASSIFICATION METRICS (CSI AND F1, WHERE HIGHER VALUES INDICATE BETTER PERFORMANCE). RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE RANDOM SEEDS. IN EACH SETTING, THE BEST AND SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

| Metric | Lead time | w/o Sat. | w/o Dec. | w/o LN | w/o Curr. | w/o HP | w/o HP+Under | w/o HP+Over | **MULTIAIR (Proposed)** |
|---|---|---|---|---|---|---|---|---|---|
| CSI score Sev. 2 or worse ($> 15\mu g/m^3$) | 1 | 0.821 ± 0.001 | 0.819 ± 0.000 | 0.821 ± 0.001 | 0.816 ± 0.000 | 0.818 ± 0.001 | 0.801 ± 0.002 | 0.815 ± 0.001 | **0.822** ± 0.000 |
| | 2 | 0.786 ± 0.001 | 0.781 ± 0.001 | 0.786 ± 0.001 | 0.780 ± 0.000 | 0.783 ± 0.002 | 0.760 ± 0.003 | 0.781 ± 0.001 | **0.786** ± 0.000 |
| | 3 | 0.757 ± 0.002 | 0.752 ± 0.002 | **0.758** ± 0.001 | 0.751 ± 0.001 | 0.755 ± 0.002 | 0.729 ± 0.003 | 0.753 ± 0.002 | 0.757 ± 0.000 |
| | 4 | 0.733 ± 0.002 | 0.726 ± 0.003 | **0.734** ± 0.001 | 0.726 ± 0.001 | 0.731 ± 0.003 | 0.704 ± 0.003 | 0.729 ± 0.003 | 0.732 ± 0.001 |
| | 5 | 0.712 ± 0.003 | 0.702 ± 0.003 | **0.713** ± 0.002 | 0.704 ± 0.001 | 0.709 ± 0.004 | 0.682 ± 0.003 | 0.707 ± 0.004 | 0.709 ± 0.001 |
| | 6 | 0.693 ± 0.003 | 0.681 ± 0.004 | **0.693** ± 0.002 | 0.684 ± 0.002 | 0.690 ± 0.005 | 0.661 ± 0.005 | 0.688 ± 0.005 | 0.689 ± 0.001 |
| | Avg. | 0.751 ± 0.002 | 0.743 ± 0.002 | **0.751** ± 0.001 | 0.743 ± 0.001 | 0.748 ± 0.003 | 0.723 ± 0.003 | 0.746 ± 0.002 | 0.749 ± 0.000 |
| F1 score Sev. 2 or worse ($> 15\mu g/m^3$) | 1 | 0.902 ± 0.000 | 0.900 ± 0.000 | 0.901 ± 0.001 | 0.899 ± 0.000 | 0.900 ± 0.001 | 0.889 ± 0.001 | 0.898 ± 0.001 | **0.902** ± 0.000 |
| | 2 | 0.880 ± 0.001 | 0.877 ± 0.001 | 0.880 ± 0.000 | 0.876 ± 0.000 | 0.878 ± 0.001 | 0.864 ± 0.002 | 0.877 ± 0.001 | **0.880** ± 0.000 |
| | 3 | 0.862 ± 0.001 | 0.858 ± 0.001 | **0.862** ± 0.001 | 0.858 ± 0.000 | 0.861 ± 0.001 | 0.843 ± 0.002 | 0.859 ± 0.001 | 0.862 ± 0.000 |
| | 4 | 0.846 ± 0.002 | 0.841 ± 0.002 | **0.847** ± 0.001 | 0.841 ± 0.001 | 0.845 ± 0.002 | 0.826 ± 0.002 | 0.843 ± 0.002 | 0.845 ± 0.000 |
| | 5 | 0.832 ± 0.002 | 0.825 ± 0.002 | **0.833** ± 0.001 | 0.826 ± 0.001 | 0.830 ± 0.003 | 0.811 ± 0.002 | 0.829 ± 0.003 | 0.830 ± 0.001 |
| | 6 | 0.819 ± 0.002 | 0.810 ± 0.003 | **0.819** ± 0.001 | 0.812 ± 0.001 | 0.816 ± 0.003 | 0.796 ± 0.004 | 0.815 ± 0.003 | 0.816 ± 0.001 |
| | Avg. | 0.857 ± 0.001 | 0.852 ± 0.002 | **0.857** ± 0.001 | 0.852 ± 0.001 | 0.855 ± 0.002 | 0.838 ± 0.002 | 0.854 ± 0.002 | 0.856 ± 0.000 |
| CSI score Sev. 3 or worse ($> 35\mu g/m^3$) | 1 | 0.755 ± 0.002 | 0.749 ± 0.001 | 0.753 ± 0.001 | 0.746 ± 0.001 | 0.749 ± 0.001 | 0.720 ± 0.002 | 0.733 ± 0.002 | **0.755** ± 0.001 |
| | 2 | 0.683 ± 0.003 | 0.673 ± 0.002 | 0.682 ± 0.003 | 0.671 ± 0.002 | 0.676 ± 0.001 | 0.635 ± 0.002 | 0.657 ± 0.003 | **0.684** ± 0.002 |
| | 3 | 0.630 ± 0.004 | 0.620 ± 0.002 | 0.629 ± 0.002 | 0.618 ± 0.003 | 0.623 ± 0.001 | 0.572 ± 0.002 | 0.603 ± 0.004 | **0.631** ± 0.002 |
| | 4 | 0.586 ± 0.005 | 0.577 ± 0.002 | 0.586 ± 0.002 | 0.577 ± 0.002 | 0.580 ± 0.003 | 0.528 ± 0.002 | 0.560 ± 0.004 | **0.588** ± 0.002 |
| | 5 | 0.549 ± 0.007 | 0.539 ± 0.003 | 0.549 ± 0.004 | 0.541 ± 0.002 | 0.541 ± 0.004 | 0.494 ± 0.004 | 0.524 ± 0.004 | **0.551** ± 0.002 |
| | 6 | 0.517 ± 0.007 | 0.507 ± 0.004 | 0.517 ± 0.007 | 0.510 ± 0.002 | 0.506 ± 0.004 | 0.466 ± 0.006 | 0.491 ± 0.005 | **0.519** ± 0.001 |
| | Avg. | 0.620 ± 0.005 | 0.611 ± 0.002 | 0.619 ± 0.001 | 0.610 ± 0.001 | 0.612 ± 0.002 | 0.569 ± 0.003 | 0.594 ± 0.004 | **0.621** ± 0.001 |
| F1 score Sev. 3 or worse ($> 35\mu g/m^3$) | 1 | 0.860 ± 0.001 | 0.856 ± 0.000 | 0.859 ± 0.001 | 0.854 ± 0.001 | 0.857 ± 0.000 | 0.837 ± 0.002 | 0.846 ± 0.001 | **0.861** ± 0.001 |
| | 2 | 0.812 ± 0.002 | 0.804 ± 0.001 | 0.811 ± 0.002 | 0.803 ± 0.002 | 0.807 ± 0.001 | 0.777 ± 0.002 | 0.793 ± 0.002 | **0.812** ± 0.001 |
| | 3 | 0.773 ± 0.003 | 0.766 ± 0.001 | 0.772 ± 0.001 | 0.764 ± 0.002 | 0.768 ± 0.001 | 0.728 ± 0.002 | 0.752 ± 0.003 | **0.774** ± 0.001 |
| | 4 | 0.739 ± 0.004 | 0.732 ± 0.001 | 0.739 ± 0.002 | 0.731 ± 0.002 | 0.734 ± 0.002 | 0.691 ± 0.002 | 0.718 ± 0.003 | **0.741** ± 0.001 |
| | 5 | 0.709 ± 0.006 | 0.700 ± 0.002 | 0.709 ± 0.004 | 0.702 ± 0.002 | 0.702 ± 0.003 | 0.661 ± 0.004 | 0.687 ± 0.004 | **0.711** ± 0.001 |
| | 6 | 0.682 ± 0.006 | 0.673 ± 0.003 | 0.681 ± 0.006 | 0.675 ± 0.002 | 0.672 ± 0.004 | 0.636 ± 0.006 | 0.659 ± 0.005 | **0.683** ± 0.001 |
| | Avg. | 0.762 ± 0.004 | 0.755 ± 0.002 | 0.762 ± 0.001 | 0.755 ± 0.001 | 0.757 ± 0.002 | 0.722 ± 0.003 | 0.742 ± 0.003 | **0.764** ± 0.001 |
| CSI score Sev. 4 ($> 75\mu g/m^3$) | 1 | 0.694 ± 0.004 | 0.682 ± 0.004 | 0.682 ± 0.005 | 0.681 ± 0.004 | 0.671 ± 0.007 | 0.595 ± 0.011 | 0.590 ± 0.010 | **0.696** ± 0.005 |
| | 2 | 0.592 ± 0.005 | 0.572 ± 0.004 | 0.574 ± 0.008 | 0.574 ± 0.006 | 0.557 ± 0.003 | 0.462 ± 0.013 | 0.460 ± 0.008 | **0.593** ± 0.008 |
| | 3 | 0.512 ± 0.008 | 0.498 ± 0.005 | 0.490 ± 0.013 | 0.495 ± 0.005 | 0.471 ± 0.007 | 0.366 ± 0.016 | 0.383 ± 0.009 | **0.518** ± 0.009 |
| | 4 | 0.446 ± 0.012 | 0.439 ± 0.006 | 0.422 ± 0.020 | 0.433 ± 0.005 | 0.402 ± 0.014 | 0.304 ± 0.018 | 0.329 ± 0.008 | **0.460** ± 0.011 |
| | 5 | 0.391 ± 0.016 | 0.385 ± 0.010 | 0.369 ± 0.021 | 0.377 ± 0.006 | 0.339 ± 0.016 | 0.259 ± 0.017 | 0.286 ± 0.010 | **0.407** ± 0.012 |
| | 6 | 0.344 ± 0.018 | 0.332 ± 0.016 | 0.323 ± 0.023 | 0.326 ± 0.007 | 0.282 ± 0.016 | 0.226 ± 0.016 | 0.248 ± 0.012 | **0.358** ± 0.011 |
| | Avg. | 0.497 ± 0.010 | 0.485 ± 0.006 | 0.476 ± 0.014 | 0.481 ± 0.005 | 0.453 ± 0.009 | 0.369 ± 0.014 | 0.383 ± 0.008 | **0.505** ± 0.009 |
| F1 score Sev. 4 ($> 75\mu g/m^3$) | 1 | 0.820 ± 0.003 | 0.811 ± 0.003 | 0.811 ± 0.003 | 0.810 ± 0.003 | 0.803 ± 0.005 | 0.746 ± 0.009 | 0.742 ± 0.008 | **0.821** ± 0.004 |
| | 2 | 0.744 ± 0.004 | 0.728 ± 0.003 | 0.729 ± 0.007 | 0.729 ± 0.005 | 0.716 ± 0.002 | 0.632 ± 0.012 | 0.630 ± 0.007 | **0.745** ± 0.006 |
| | 3 | 0.677 ± 0.007 | 0.665 ± 0.004 | 0.657 ± 0.012 | 0.663 ± 0.004 | 0.640 ± 0.006 | 0.536 ± 0.017 | 0.554 ± 0.009 | **0.683** ± 0.008 |
| | 4 | 0.617 ± 0.011 | 0.611 ± 0.006 | 0.594 ± 0.020 | 0.604 ± 0.005 | 0.573 ± 0.014 | 0.466 ± 0.021 | 0.495 ± 0.009 | **0.630** ± 0.011 |
| | 5 | 0.562 ± 0.016 | 0.556 ± 0.010 | 0.539 ± 0.023 | 0.547 ± 0.006 | 0.506 ± 0.018 | 0.411 ± 0.021 | 0.445 ± 0.012 | **0.579** ± 0.012 |
| | 6 | 0.511 ± 0.020 | 0.499 ± 0.018 | 0.488 ± 0.026 | 0.492 ± 0.008 | 0.439 ± 0.019 | 0.368 ± 0.021 | 0.397 ± 0.016 | **0.528** ± 0.012 |
| | Avg. | 0.655 ± 0.010 | 0.645 ± 0.007 | 0.636 ± 0.014 | 0.641 ± 0.005 | 0.613 ± 0.010 | 0.526 ± 0.016 | 0.544 ± 0.009 | **0.664** ± 0.009 |

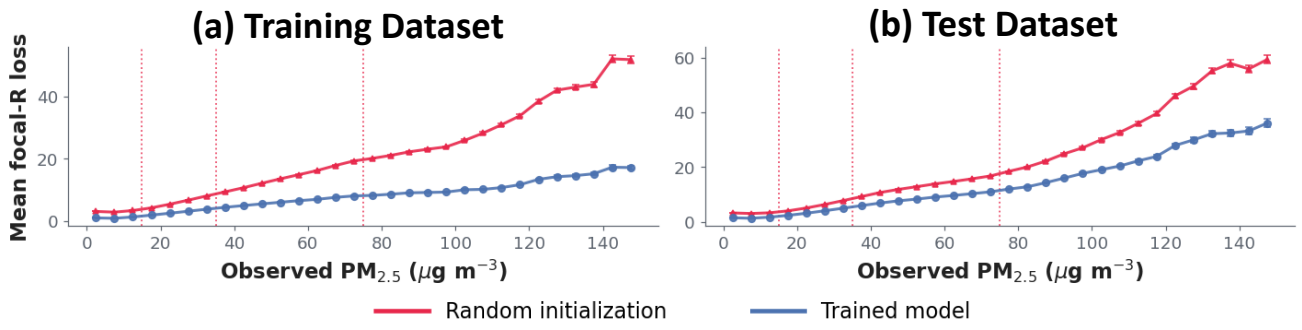



Fig. 5. Mean focal-R loss across observed $PM_{2.5}$ concentrations on the (a) training and (b) test sets. Red triangles and blue circles represent MULTIAIR at random initialization and after training, respectively. Vertical dotted lines indicate the severity boundaries at 15, 35, and 75 $\mu g/m^3$.

TABLE VIII

ABLATION STUDY RESULTS BASED ON REGRESSION METRICS (RMSE AND MAE, WHERE LOWER VALUES INDICATE BETTER PERFORMANCE). RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE RANDOM SEEDS. IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

| Metric | Lead time | w/o Sat. | w/o Dec. | w/o LN | w/o Curr. | w/o HP | w/o HP+Under | w/o HP+Over | **MULTIAIR (Proposed)** |
|---|---|---|---|---|---|---|---|---|---|
| RMSE | 1 | 4.014 ± 0.016 | 4.106 ± 0.009 | 4.063 ± 0.024 | 4.167 ± 0.017 | 4.133 ± 0.036 | 4.635 ± 0.006 | 4.487 ± 0.050 | **4.008** ± 0.013 |
| | 2 | 5.217 ± 0.022 | 5.361 ± 0.035 | 5.225 ± 0.027 | 5.389 ± 0.014 | 5.358 ± 0.039 | 6.055 ± 0.036 | 5.776 ± 0.047 | **5.201** ± 0.027 |
| | 3 | 6.160 ± 0.024 | 6.307 ± 0.057 | 6.144 ± 0.024 | 6.329 ± 0.029 | 6.306 ± 0.044 | 7.133 ± 0.079 | 6.708 ± 0.046 | **6.134** ± 0.033 |
| | 4 | 6.964 ± 0.030 | 7.109 ± 0.076 | **6.918** ± 0.038 | 7.137 ± 0.054 | 7.087 ± 0.040 | 7.993 ± 0.136 | 7.457 ± 0.063 | 6.920 ± 0.041 |
| | 5 | 7.673 ± 0.046 | 7.841 ± 0.103 | **7.576** ± 0.046 | 7.854 ± 0.085 | 7.749 ± 0.025 | 8.710 ± 0.180 | 8.093 ± 0.095 | 7.613 ± 0.050 |
| | 6 | 8.287 ± 0.062 | 8.488 ± 0.126 | **8.171** ± 0.038 | 8.471 ± 0.115 | 8.333 ± 0.021 | 9.296 ± 0.194 | 8.656 ± 0.145 | 8.231 ± 0.067 |
| | Avg. | 6.386 ± 0.033 | 6.535 ± 0.067 | **6.349** ± 0.016 | 6.558 ± 0.048 | 6.494 ± 0.026 | 7.304 ± 0.103 | 6.863 ± 0.074 | 6.351 ± 0.037 |
| MAE | 1 | 2.734 ± 0.011 | 2.782 ± 0.008 | 2.769 ± 0.026 | 2.829 ± 0.009 | 2.820 ± 0.021 | 3.175 ± 0.021 | 2.967 ± 0.014 | **2.728** ± 0.007 |
| | 2 | 3.458 ± 0.015 | 3.543 ± 0.026 | 3.468 ± 0.022 | 3.567 ± 0.004 | 3.546 ± 0.028 | 4.028 ± 0.059 | 3.713 ± 0.017 | **3.445** ± 0.011 |
| | 3 | 4.017 ± 0.019 | 4.118 ± 0.039 | **3.999** ± 0.026 | 4.129 ± 0.012 | 4.103 ± 0.034 | 4.689 ± 0.083 | 4.273 ± 0.022 | 4.000 ± 0.014 |
| | 4 | 4.495 ± 0.020 | 4.618 ± 0.052 | **4.444** ± 0.025 | 4.610 ± 0.029 | 4.574 ± 0.036 | 5.213 ± 0.102 | 4.745 ± 0.034 | 4.474 ± 0.017 |
| | 5 | 4.922 ± 0.026 | 5.077 ± 0.066 | **4.838** ± 0.030 | 5.039 ± 0.048 | 4.990 ± 0.036 | 5.656 ± 0.115 | 5.161 ± 0.053 | 4.898 ± 0.018 |
| | 6 | 5.297 ± 0.033 | 5.482 ± 0.081 | **5.206** ± 0.035 | 5.416 ± 0.062 | 5.365 ± 0.038 | 6.042 ± 0.124 | 5.536 ± 0.075 | 5.277 ± 0.024 |
| | Avg. | 4.154 ± 0.020 | 4.270 ± 0.045 | **4.121** ± 0.022 | 4.265 ± 0.024 | 4.233 ± 0.031 | 4.801 ± 0.083 | 4.399 ± 0.035 | 4.137 ± 0.015 |

TABLE IX

ABLATION STUDY RESULTS IN TERMS OF EXTRA METRICS. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE RANDOM SEEDS. FOR METRICS WITH AN UPWARD ARROW (↑), HIGHER VALUES INDICATE BETTER PERFORMANCE, WHEREAS FOR THOSE WITH A DOWNWARD ARROW (↓), LOWER VALUES ARE PREFERABLE. IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

| Metric | | w/o Sat. | w/o Dec. | w/o LN | w/o Curr. | w/o HP | w/o HP+Under | w/o HP+Over | **MULTIAIR (Proposed)** |
|---|---|---|---|---|---|---|---|---|---|
| **Class.** | ACC (↑) | 0.809 ± 0.001 | 0.803 ± 0.002 | 0.810 ± 0.002 | 0.804 ± 0.001 | 0.805 ± 0.002 | 0.779 ± 0.005 | 0.799 ± 0.002 | **0.810** ± 0.001 |
| | POD (↑) | **0.737** ± 0.012 | 0.728 ± 0.005 | 0.723 ± 0.009 | 0.724 ± 0.012 | 0.726 ± 0.009 | 0.703 ± 0.004 | 0.716 ± 0.005 | 0.733 ± 0.006 |
| | FAR (↓) | 0.208 ± 0.007 | 0.214 ± 0.007 | **0.192** ± 0.011 | 0.210 ± 0.015 | 0.207 ± 0.008 | 0.257 ± 0.008 | 0.227 ± 0.001 | 0.200 ± 0.008 |
| | F1 (↑) | 0.763 ± 0.004 | 0.756 ± 0.002 | 0.763 ± 0.001 | 0.756 ± 0.001 | 0.758 ± 0.001 | 0.722 ± 0.003 | 0.743 ± 0.003 | **0.765** ± 0.001 |
| **Reg.** | RMSE (↓) | 6.548 ± 0.035 | 6.700 ± 0.073 | **6.500** ± 0.020 | 6.718 ± 0.055 | 6.648 ± 0.024 | 7.474 ± 0.115 | 7.005 ± 0.078 | 6.510 ± 0.040 |
| | NME (↓) | 22.330 ± 0.109 | 22.955 ± 0.241 | **22.152** ± 0.118 | 22.928 ± 0.128 | 22.756 ± 0.166 | 25.807 ± 0.448 | 23.650 ± 0.189 | 22.240 ± 0.080 |
| | $\rho$ (↑) | 0.898 ± 0.001 | 0.893 ± 0.002 | 0.899 ± 0.001 | 0.892 ± 0.002 | 0.893 ± 0.001 | 0.864 ± 0.004 | 0.881 ± 0.003 | **0.900** ± 0.002 |

TABLE X

HYPERPARAMETER SENSITIVITY TEST RESULTS IN TERMS OF EXTRA METRICS. FOR METRICS WITH AN UPWARD ARROW (↑), HIGHER VALUES CORRESPOND TO BETTER PERFORMANCE, WHEREAS FOR THOSE WITH A DOWNWARD ARROW (↓), LOWER VALUES ARE PREFERABLE. IN EACH SETTING, THE BEST PERFORMANCE IS IN **BOLD**.

| **Metric** | | Input Length ($T_t$) | | | | Hidden Dimension ($d_h$) | | | Gamma ($\gamma$) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 4 | 7 | 13 | 25 | 64 | 128 | 256 | 1 | 2 | 3 |
| **Class.** | ACC (↑) | 0.801 | **0.812** | 0.811 | **0.812** | 0.805 | **0.812** | 0.801 | 0.806 | **0.812** | 0.808 |
| | POD (↑) | 0.730 | 0.729 | 0.727 | **0.737** | 0.728 | 0.729 | **0.736** | **0.748** | 0.729 | 0.728 |
| | FAR (↓) | 0.219 | **0.192** | 0.197 | 0.194 | 0.207 | **0.192** | 0.228 | 0.220 | **0.192** | 0.205 |
| | F1 (↑) | 0.754 | 0.766 | 0.763 | **0.770** | 0.759 | **0.766** | 0.754 | 0.764 | **0.766** | 0.760 |
| **Reg.** | RMSE (↓) | 6.712 | 6.444 | 6.463 | **6.346** | 6.672 | **6.444** | 6.755 | 6.571 | **6.444** | 6.539 |
| | NME (↓) | 35.747 | **34.225** | 34.716 | 35.765 | 35.383 | **34.225** | 36.460 | 35.443 | **34.225** | 34.617 |
| | $\rho$ (↑) | 0.893 | 0.901 | 0.901 | **0.905** | 0.894 | **0.901** | 0.891 | 0.897 | **0.901** | 0.898 |